Natural Language Processing

Instructor

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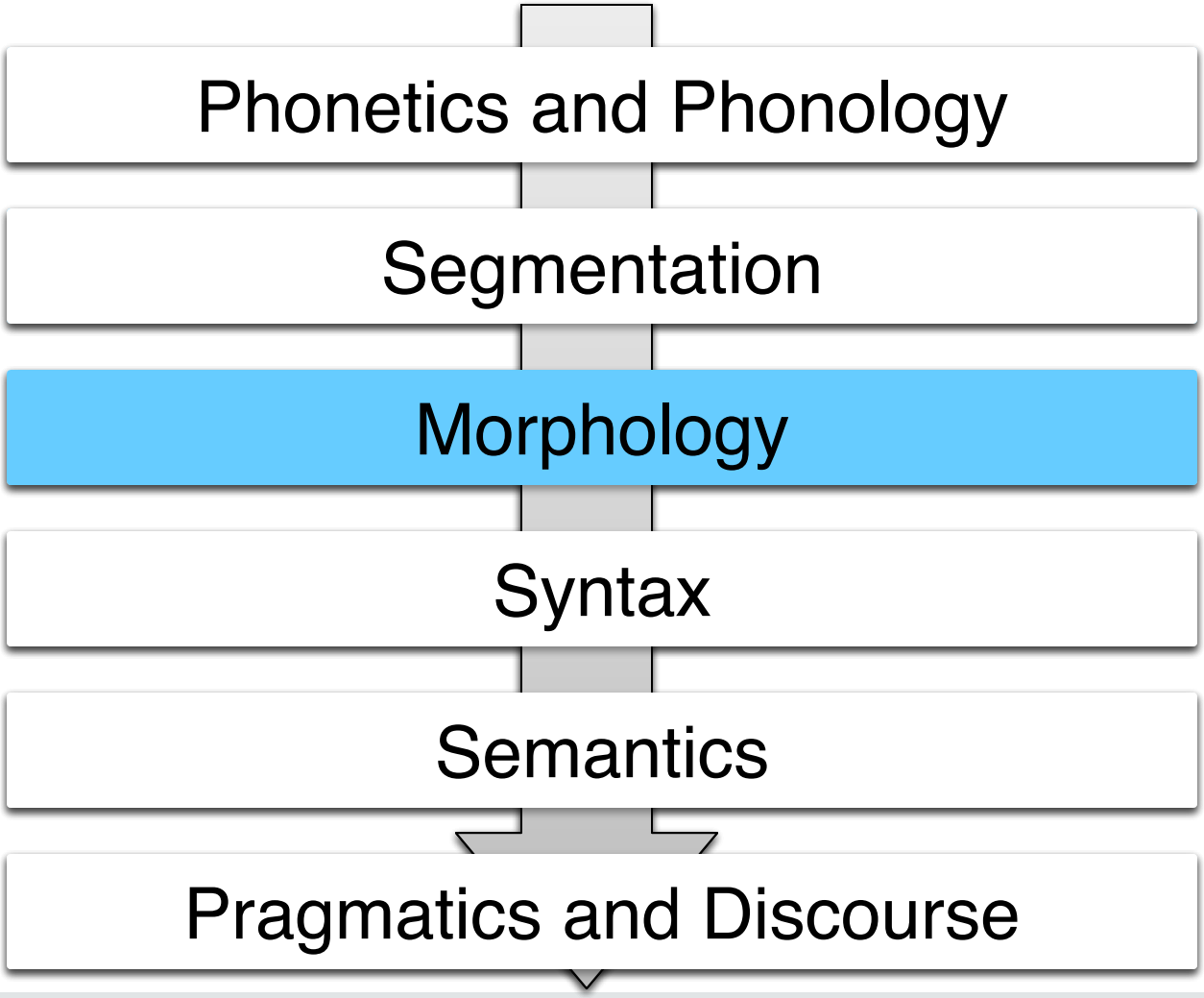
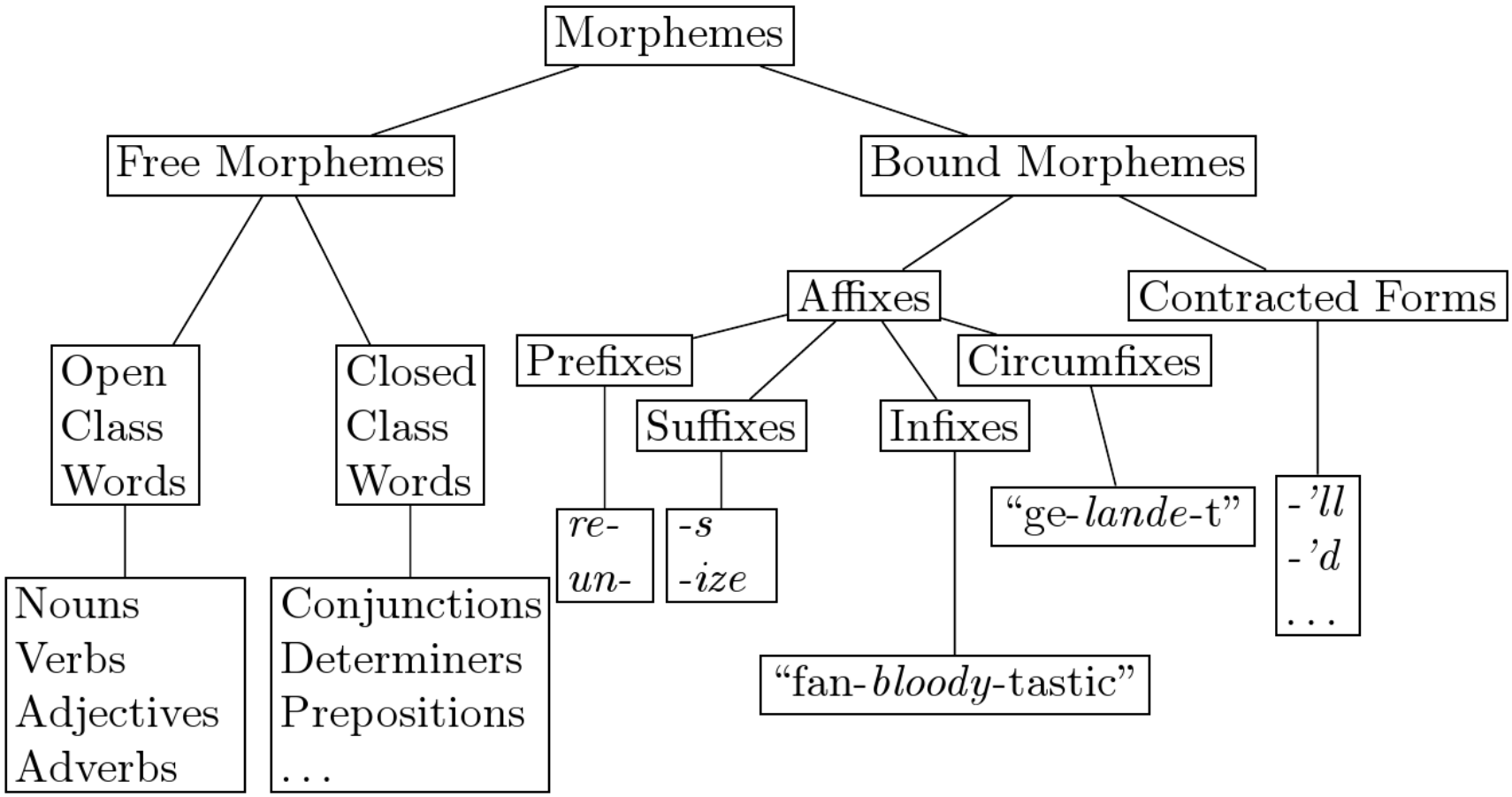
Analysis Levels in Language Understanding



Classification of Morphemes



Morphological Normalization

- **Morphological normalization** consists in identifying a single canonical representative for morphologically related word-forms

Methods

- Stemming
- Lemmatization

What is Stemming?

Stemming is an algorithmic approach to strip off the endings of words

sitting → sitt

anarchism, anarchistic → anarchi

Objective: group words belonging to the same morphological family by transforming them into the same stemmed representation

- the stems obtained do not necessarily correspond to a real word form

Well-known stemming algorithms for English have been developed by Lovins (1968) and Porter (1980)

Rule- Based Stemming

Rule-based stemming methods are hard to create, but can be executed very quickly at runtime.

Stemming- Porter's Stemmer

- Based on a limited set of general **cascaded transformational rules**:
  - -ational → -ate, relational → relate
- Variants exist for **many languages**: English, French, Spanish, Portuguese, Italian, Romanian, German, Dutch, Swedish, Norwegian, Danish, Russian, Finnish, Hungarian, Turkish
- **Fast**
- Accurate enough for **some** applications, e.g. Information Retrieval
- Official Link- https://tartarus.org/martin/PorterStemmer/index.html

Stemming- Porter's Stemmer

- Step 1a
  - SSES → SS
    caresses → caress
- Step 1b
  - (m>0) EED → EE
    agreed → agree
- Step 1c
  - (*v*) Y → I
    happy → happi
- Step 2
  - (m>0) ATIONAL → ATE
    relational → relate
- Step 3
  - (m>0) ICATE → IC
    triplicate → triplic
- Step 4
  - (m>1) AL →
    revival → reviv
- Step 5a
  - (m>1) E →
    probate → probat
- Step 5b
  - (m > 1 and *d and *L) → single letter
    controll → control

Stemming- Porter's Stemmer

Original Word	**Stemmed Word**
vision	vision
visible	visibl
visibility	visibl
visionary	visionari
visioner	vision
visual	visual

Comparison of Stemmers

Sample text: Such an analysis can reveal features that are not easily visible from the variations in the individual genes and can lead to a picture of expression that is more biologically transparent and accessible to interpretation

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Issues with Stemming

- **Under-stemming:**
  - running → run
  - runner → runner

- **Over-stemming:**
  - appendicitis → append
  - append → append

What is Lemmatization?

- The task of **lemmatization** is to “undo” the inflectional changes which a baseform undergoes
  - cats → cat
  - am, is, are → be

- In order to disambiguate ambiguous cases, lemmatization is usually combined with **part-of-speech tagging**
  - left → leave (verb)
  - left → left (adjective)

Lemmatization- General Approaches

- **Simplest solution :** Have a list of all possible word forms and associated lemma information
- **Brute force:** try every possible segmentation of word and see which ones match known stems and affixes
- **Rule-based**
  - Have list of known affixes, see which ones apply
  - Have knowledge about allowable combinations

Stemming vs Lemmatization

Original	Stemmed	Lemmatized
visibilities	visibl	visibility
adhere	adher	adhere
adhesion	adhes	adhesion
appendicitis	append	appendicitis
oxen	oxen	ox
indices	indic	index
swum	swum	swim

Lemmatization- Available Packages (Python)

- Wordnet Lemmatizer
- Spacy Lemmatizer
- TextBlob
- Stanford CoreNLP
- Gensim Lemmatizer
- …..

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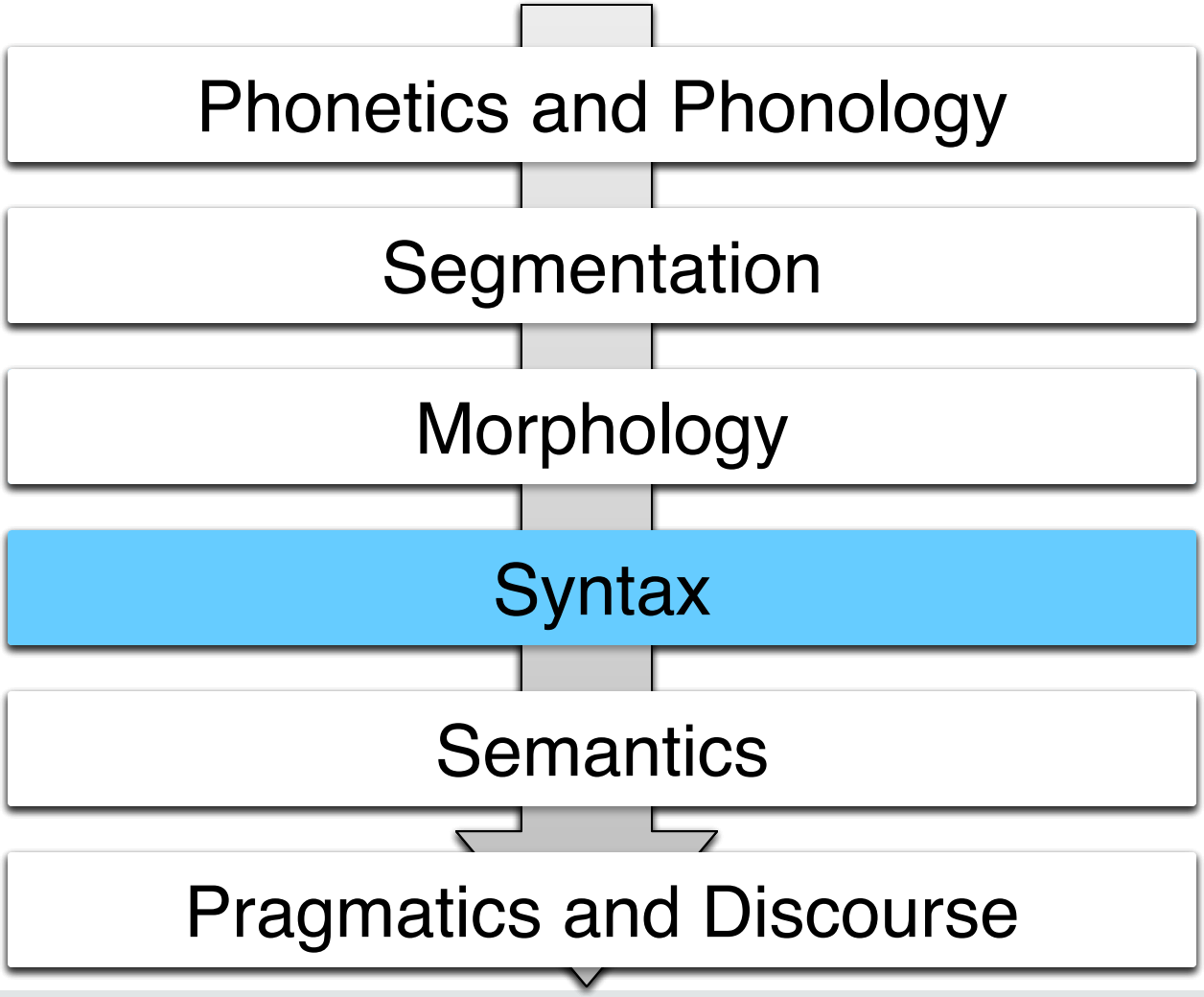

What is Syntax?

- **Syntax** refers to the way words are arranged together

- "Syntax is the study of the regularities and constraints of word order and phrase structure"

  (Manning & Schütze)

Why is Syntax necessary?

- Number of ways in which words can be arranged together to form sentences — —- A lot!!
  - Yet, we can understand sentences we have never heard or read before
- Rules of syntax specify the possible organization of words in sentences and allows us to determine sentence's structure(s)

What is Parts of Speech ?

- Linguists classify words on the basis of certain features these words have in common.
  - The classes that we use are called ***Parts of Speech*** (or *lexical categories*, or *word classes*).

Parts of Speech

- Traditional English Parts of Speech

N	Noun	*chair, bandwidth, pacing*
V	Verb	*study, debate, munch*
ADJ	Adjective	*purple, tall, ridiculous*
ADV	Adverb	*unfortunately, slowly*
P	Preposition	*of, by, to*
PRO	Pronoun	*I, me, mine*
DET	Determiner	*the, a, that, those*

Penn Treebank Tagset

1. CC Coord. conjunc.
2. CD Cardinal number
3. DT Determiner
4. EX Existential there
5. FW Foreign word
6. IN Prep./subord. conj.
7. JJ Adject.
8. JJR Adject., comp.
9. JJS Adject., superl.
10. LS List item marker
11. MD Modal
12. NN Noun, sing. or mass
13. NNS Noun, plural
14. NNP Proper noun, sing.
15. NNPS Proper noun, plural
16. PDT Predeterminer
17. POS Possessive ending
18. PRP Personal pronoun
19. PP$ Poss. pronoun
20. RB Adverb
21. RBR Adverb, comp.
22. RBS Adverb, superl.
23. RP Particle
24. SYM Symbol
25. TO to
26. UH Interjection
27. VB V, base form
28. VBD V, past tense
29. VBG V, gerund/pres. part.
30. VBN V, past part.
31. VBP V, non-3rd ps. sing. pres.
32. VBZ V, 3rd ps. sing. pres.
33. WDT wh-det.
34. WP wh-pronoun
35. WP$ Poss. wh-pronoun
36. WRB wh-adverb
37. # Pound sign
38. $ Dollar sign
39. . Sent.-final punct.
40. , Comma
41. : Colon, semi-colon
42. (L. bracket char.
43.) R. bracket char.
44. “ Straight dbl. quote
45. ‘ L. open sngl. quote
46. “ L. open dbl. quote
47. ’ R. close sngl. quote
48. ” R. close dbl. quote

Parts of Speech Tagging: Ambiguities

Problem: Multiple possibilities for assigning parts of speech

- *The* <u>*back*</u> *door* = ADJ
- *On my* <u>*back*</u> = N
- *Win the voters* <u>*back*</u> = ADV
- Promised to <u>*back*</u> the bill = V